

application modes and the next generation of small sensor form factors. It is highly configurable and suitable for through-beam retro-reflective and diffuse (proximity sensing) photoelectric sensor applications.

Triad Semiconductor
<https://triadsemi.com>

SoC for IoT devices

Silicon Labs BG27 is a new addition to company's xG27 system on chip (SoC), which enables secure, energy-friendly multiprotocol wireless networking for IoT devices. The BG27 comes in a



compact packaging measuring just 2.6mm × 2.3mm. It can enable medical device makers to integrate Bluetooth LE connectivity on devices with small form factor, such as the smart tooth implant. The BG27 enables device makers to add high performance and secure low-power Bluetooth LE on tiny form factor devices, expanding the possibilities of smart connected medical devices and wearables. The SoC features an integrated DC-DC boost that allows operation down to 0.8 volts, thus enabling the use of single-cell alkaline, silver oxide, and 1.55V button cell batteries typically used in medical applications such as battery-operated patches, wearable electrocardiograms, and continuous glucose monitors.

Silicon Labs
<https://www.silabs.com>

Tiny ultra-fast servo drives

Celera Motion's Denali series is a line of ultra-fast, compact servo drives suitable for extremely sensitive medical

and industrial applications. The servo drives can work in the 250W power range and are designed for surgical robotics, end-effectors, haptic devices, small joints, and other compact robotics applications. The drives are the world's smallest, weighing just



8 grams. The Denali series features EtherCAT and CANopen communication protocols, optimised for demanding multi-axis applications. It supports EtherCAT with a bus latency down to 1 cycle, which improves the cost-efficiency of embedding multiple axes into a single PCB. The servo drive will reduce the cost, size, and time of manufacturing. The high efficiency and low standby power consumption of device make it suitable for use in service robots, surgical robots, industrial grippers, and lab automation applications.

Celera Motion
<https://www.celeramotion.com>

Single-chip BMS controllers

Qorvo PAC22140 and PAC25140 are the industry's first single-chip power application controllers to offer support for battery packs with up to 20 cells in series. The highly integrated controllers save designers more than 50% of PCB space and lower the material cost by 30%, while reducing time to market, leveraging Qorvo's hardware and firmware ecosystem.

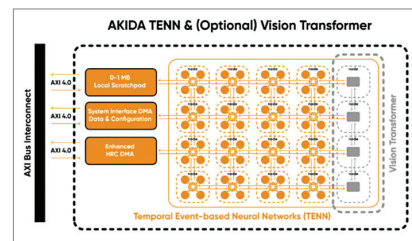
The devices integrate all essential analogue and power management peripherals with a 32-bit Arm Cortex M0 or M4F microcontroller and include cell balancing, monitoring, and protection for 10- to 20-cell battery packs. They leverage an intelligent motor control foundation to enable intelligent battery management solutions (BMS) for a wide range of industrial, e-mobility, and battery backup applications.

Qorvo
<https://www.qorvo.com>

Powerful neural processing system

BrainChip's second-generation Akida platform is a powerful neural processing system, architected for embedded edge AI applications. The 2nd Gen Akida platform offers efficient 8-bit processing suitable for time domain convolutions and vision transformer acceleration, for high level of perfor-

mance in sub-watt devices. It includes temporal event based neural nets spatial-temporal convolutions that supercharge the processing of raw time-continuous streaming data, such as



video analytics, target tracking, audio classification, analysis of MRI and CT scans for vital signs prediction, and time series analytics used in forecasting and predictive maintenance. The Akida platform offers simpler implementations by consuming raw data directly from sensors, which drastically reduces model size and operations performed, while maintaining very high accuracy. This can enable design engineers to shrink design cycles and lower the cost of development.

BrainChip
<https://brainchip.com>

VPUs for intelligent cameras

The Hailo 15 family of vision processors (VPUs) will introduce the AI capabilities to cameras used in a wide range of applications for different industries.



It is designed for integration directly into intelligent cameras to deliver video processing and analytics. The Hailo 15 series offers processing power up to 20 tera operations per second, thus delivering over 5x higher performance than currently available solutions in the market, at a comparable price point. All Hailo 15 VPUs support multiple input streams at 4K resolution and combine a powerful CPU and DSP subsystems with Hailo's AI core. The Hailo 15 VPU will enable identification of smaller and more distant objects with higher accuracy and less false alarms.

Hailo
<https://hailo.ai>